B4.1 Adaptation to environment

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 DEFINE habitat. [B4.1.1] [1]

2 STATE two abiotic variables that affect the distribution of animals. [B4.1.3] [2]

3 LIST three conditions required for coral reefs to form. [B4.1.5] [3]

4 NAME the two climate factors that decide land biomes. [B4.1.6] [1]

Section B: Understanding

14 MARKS

5 EXPLAIN two adaptations of a grass species to life on sand dunes. [B4.1.2] [4]

6 EXPLAIN how mangrove trees survive in waterlogged, salty soil. [B4.1.2] [3]

7 EXPLAIN why biomes on different continents have similar communities. [B4.1.7] [3]

8 **COMPARE AND CONTRAST** adaptations to hot deserts and tropical rainforests, using named species. [4]

[B4.1.8]

HOT DESERT	TROPICAL RAINFOREST

Section C: Working with data

13 MARKS

All data in this section are invented for practice. The patterns are realistic.

9 [B4.1.4]

A transect was run from open grass into a wood. At each point, light was measured and a shade plant was counted in a quadrat.

DISTANCE / M	0	5	10	15	20	25
LIGHT / KLX	40	32	18	9	4	1
NUMBER OF PLANTS	0	1	6	14	9	2

a **DESCRIBE** how numbers of the plant change as light falls. [2]

b **ESTIMATE** the optimum light level and the range of tolerance. [2]

c **SUGGEST** a reason why the plant is rare at 25 m. [1]

d **EXPLAIN** why these data do not prove that light controls the distribution. [2]

10 [B4.1.6, B4.1.7]

Climate data for three sites are shown.

SITE	MEAN TEMPERATURE / °C	RAINFALL / MM PER YEAR
P	26	2800
Q	−10	200
R	27	100

a **IDENTIFY** the likely biome at each site. [3]

b **EXPLAIN** why trees are absent at site Q. [2]

c **SUGGEST** why few plants grow at site R. [1]

Section D: Longer answer

6 MARKS

11 **EXPLAIN** how abiotic factors determine where coral reefs can form, and how coral reefs depend on a relationship between two organisms. [B4.1.3, B4.1.5] [6]